EBS00004

# HOW TO USE THIS MANUAL

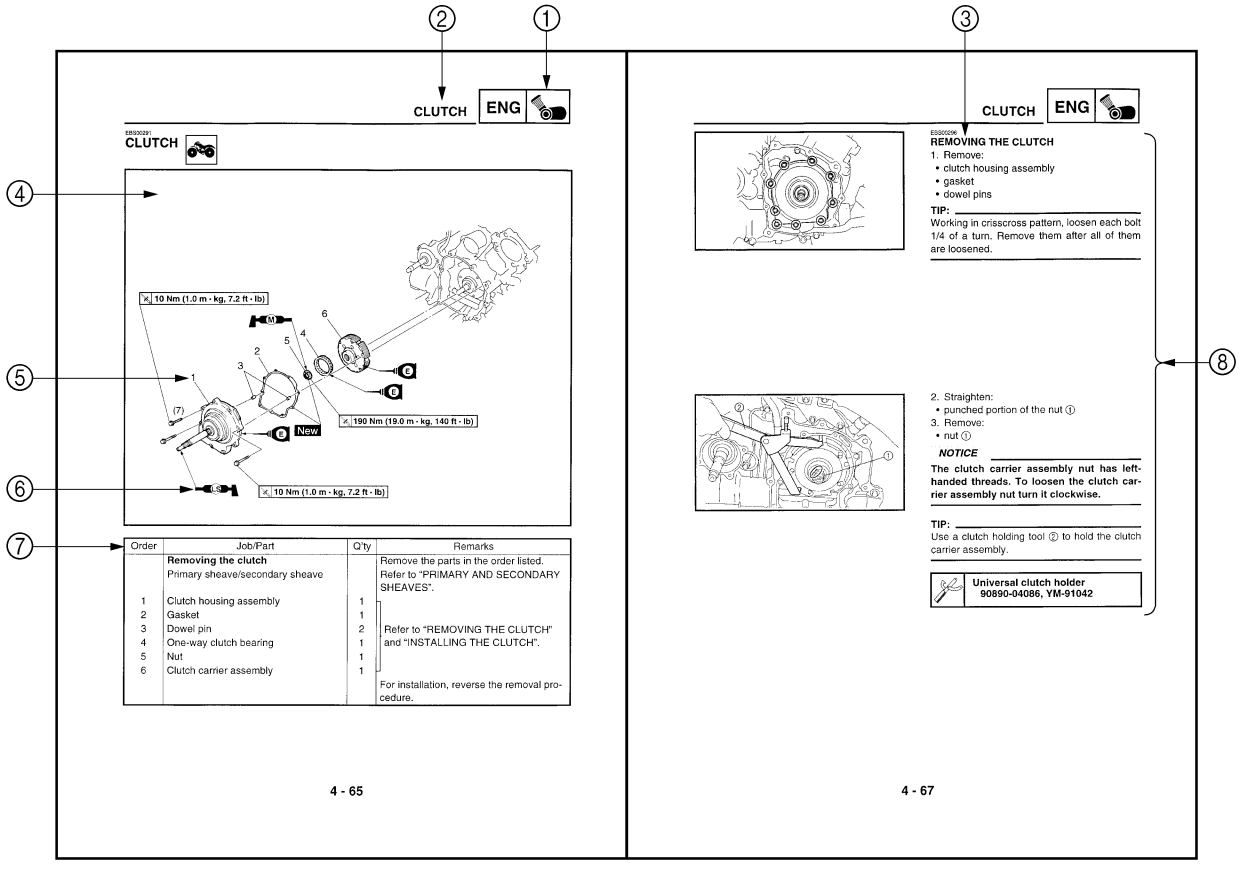
## MANUAL ORGANIZATION

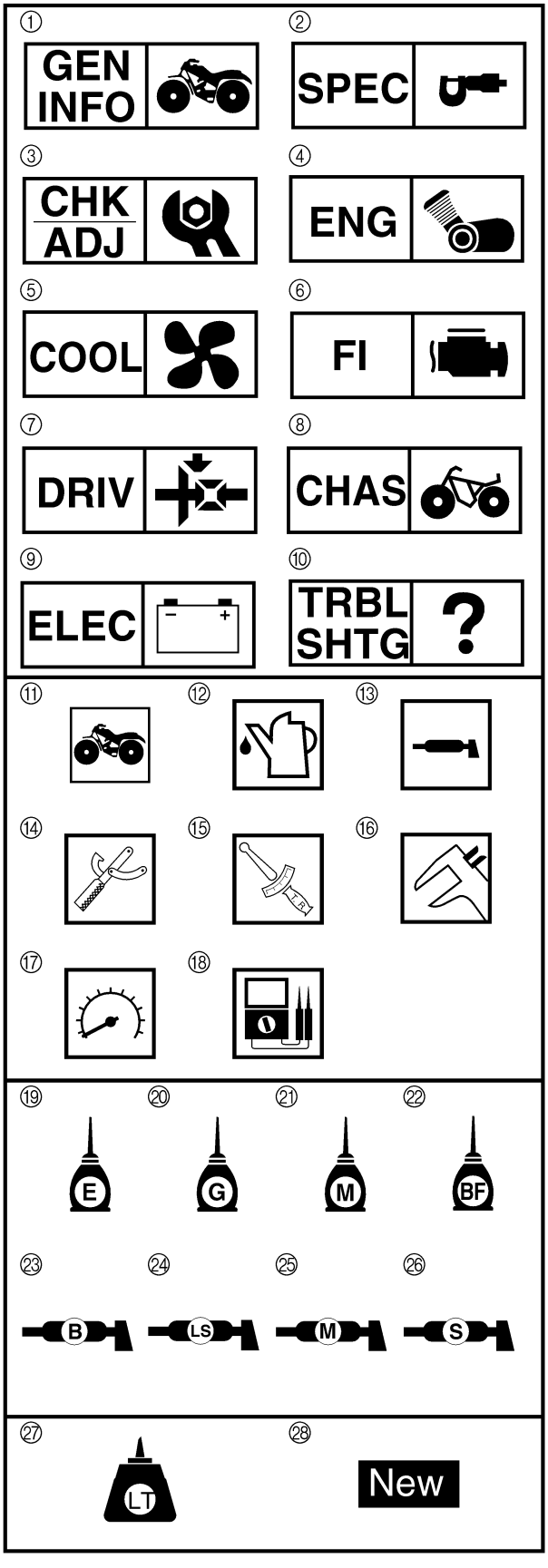
This manual consists of chapters for the main categories of subjects. (See “symbols”)
1st title ①: This is the title of the chapter with its symbol in the upper right corner of each page.
2nd title ②: This title indicates the section of the chapter and only appears on the first page of each section. It is located in the upper left corner of the page.
3rd title ③: This title indicates a sub-section that is followed by step-by-step procedures accompanied by corresponding illustrations.

## EXPLODED DIAGRAMS

To help identify parts and clarify procedure steps, there are exploded diagrams at the start of each removal and disassembly section.

1. An easy-to-see exploded diagram ④ is provided for removal and disassembly jobs.
2. Numbers ⑤ are given in the order of the jobs in the exploded diagram. A number that is enclosed by a circle indicates a disassembly step.
3. An explanation of jobs and notes is presented in an easy-to-read way by the use of symbol marks ⑥. The meanings of the symbol marks are given on the next page.
4. A job instruction chart ⑦ accompanies the exploded diagram, providing the order of jobs, names of parts, notes in jobs, etc.
5. For jobs requiring more information, the step-by-step format supplements ⑧ are given in addition to the exploded diagram and the job instruction chart.






EBS00006

## SYMBOLS

The following symbols are not relevant to every vehicle.

Symbols ① to ⑩ indicate the subject of each chapter.

① General information
② Specifications
③ Periodic checks and adjustments
④ Engine
⑤ Cooling system
⑥ Fuel injection system
⑦ Drive train
⑧ Chassis
⑨ Electrical
⑩ Troubleshooting

Symbols ⑪ to ⑱ indicate the following

⑪ Can be serviced with engine mounted
⑫ Filling fluid
⑬ Lubricant
⑭ Special tool
⑮ Torque
⑯ Wear limit, clearance
⑰ Engine speed
⑱ Electrical data (Ω, V, A)

Symbols ⑲ to ㉖ in the exploded diagrams indicate the types of lubricants and lubrication points.

⑲ Apply engine oil
⑳ Apply gear oil
㉑ Apply molybdenum disulfide oil
㉒ Apply brake fluid
㉓ Apply wheel bearing grease
㉔ Apply lithium-soap-based grease
㉕ Apply molybdenum disulfide grease
㉖ Apply silicone grease

Symbols ㉗ to ㉘ in the exploded diagrams indicate where to apply a locking agent ㉗ and when to install a new part ㉘.

㉗ Apply the locking agent (LOCTITE®)
㉘ Replace

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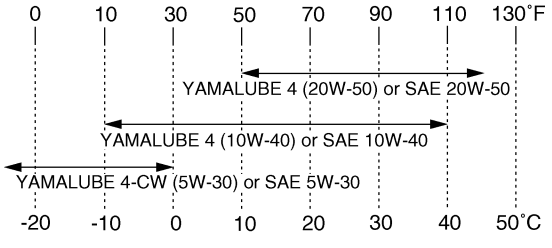
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# SPECIFICATIONS

## GENERAL SPECIFICATIONS

*1 For models equipped with panel wheels and oil damper shock absorbers
*2 For models equipped with cast wheels and oil damper shock absorbers
*3 For models equipped with cast wheels and gas-oil damper shock absorbers
*4 For models equipped with cast wheels, gas-oil damper shock absorbers, and overfenders

| Item | Standard |
|---|---|
| **Model code** | 5B41/5B45/5B48 *1<br>5B49 *2<br>5B4B *3<br>5B4E *4 |
| **Dimensions** | |
| Overall length | 2,885 mm (113.6 in) |
| Overall width | 1,385 mm (54.5 in) *1, *2, *3<br>1,516 mm (59.7 in) *4 |
| Overall height | 1,853 mm (73.0 in) *1, *2<br>1,901 mm (74.8 in) *3<br>1,865 mm (73.4 in) *4 |
| Seat height | 818 mm (32.2 in) |
| Wheelbase | 1,910 mm (75.2 in) |
| Minimum ground clearance | 280 mm (11.0 in) |
| Minimum turning radius | 3,900 mm (154 in) |
| **Basic weight** | |
| With oil and fuel | 540.0 kg (1,190 lb) *1, *2<br>548.0 kg (1,208 lb) *3, *4 |
| **Engine** | |
| Engine type | Liquid-cooled 4-stroke, SOHC |
| Cylinder arrangement | Forward-inclined single cylinder |
| Displacement | 686.0 $cm^3$ |
| Bore × stroke | 102.0 × 84.0 mm (4.02 × 3.31 in) |
| Compression ratio | 9.20 : 1 |
| Standard compression pressure (at sea level) | 450 kPa (4.50 $kg/cm^2$, 64.0 psi) |
| Starting system | Electric starter |
| **Lubrication system** | Wet sump |
| **Oil type or grade** | |
| Engine oil |  <br>API service SG type or higher,<br>JASO standard MA |
| Final gear oil | SAE 80 API GL-4 Hypoid gear oil |
| Differential gear oil | SAE 80 API GL-4 Hypoid gear oil |

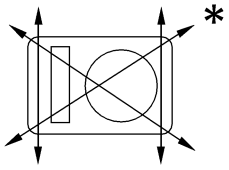
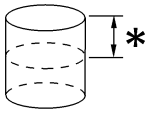
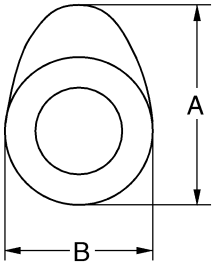
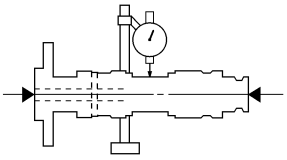
| Item | | Standard |
|---|---|---|
| **Oil quantity** | | |
| Engine oil | | |
| Without oil filter cartridge replacement | | 2.00 L (1.76 Imp qt, 2.11 US qt) |
| With oil filter cartridge replacement | | 2.10 L (1.85 Imp qt, 2.22 US qt) |
| Total amount | | 3.00 L (2.64 Imp qt, 3.17 US qt) |
| Final gear oil | | |
| Periodic oil change | | 0.25 L (0.22 Imp qt, 0.26 US qt) |
| Total amount | | 0.28 L (0.25 Imp qt, 0.30 US qt) |
| Differential gear case oil | | |
| Periodic oil change | | 0.18 L (0.16 Imp qt, 0.19 US qt) |
| Total amount | | 0.20 L (0.18 Imp qt, 0.21 US qt) |
| **Radiator capacity (including all routes)** | | 2.35 L (2.07 Imp qt, 2.48 US qt) |
| **Air filter** | | Wet type element |
| **Fuel** | | |
| Type | | Unleaded gasoline only |
| Fuel tank capacity | | 29.3 L (6.45 Imp gal, 7.74 US gal) |
| **Fuel injector** | | |
| Type/quantity | | E252019/1 |
| Manufacturer | | NIPPON INJECTOR |
| **Spark plug** | | |
| Type/manufacturer | | CPR7EA-9/NGK |
| Spark plug gap | | 0.8 ~ 0.9 mm (0.031 ~ 0.035 in) |
| **Clutch type** | | Wet, centrifugal automatic |
| **Transmission** | | |
| Primary reduction system | | V-belt |
| Secondary reduction system | | Shaft drive |
| Secondary reduction ratio | | 41/21 × 17/12 × 33/9 (10.142) |
| Transmission type | | V-belt automatic |
| Operation | | Right hand operation |
| Single speed automatic | | 2.380 ~ 0.783 : 1 |
| Sub transmission ratio | low | 31/16 (1.938) |
| | high | 31/21 (1.476) |
| Reverse gear | | 23/14 × 28/23 (2.000) |
| **Chassis** | | |
| Frame type | | Steel tube frame |
| Caster angle | | 5.0° |
| Camber angle | | 0° |
| Kingpin angle | | 12.0° |
| Kingpin offset | | 0 mm (0 in) |
| Trail | | 26.0 mm (1.02 in) |
| Tread (STD) | front | 1,115.0 mm (43.90 in) |
| Tread (STD) | rear | 1,105.0 mm (43.50 in) |
| Toe-in (with tires touching the ground) | | 15.0 ~ 25.0 mm (0.59 ~ 0.98 in) |

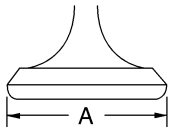
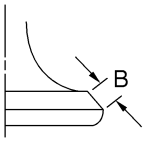
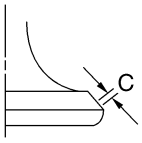
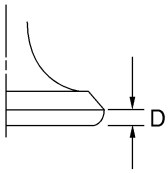
| Item | | Standard |
|---|---|---|
| **Tire** | | |
| Type | front | Tubeless |
| | rear | Tubeless |
| Size | front | 25 × 8.00-12NHS |
| | rear | 25 × 10.00-12NHS |
| Manufacturer/model | front | MAXXIS/M951Y |
| | rear | MAXXIS/M952Y |
| **Tire pressure (cold tire)** | | |
| Maximum load* | | 367.0 kg (809 lb) *1, *2<br>359.0 kg (791 lb) *3, *4 |
| Off-road riding | front | 63 ~ 77 kPa (0.63 ~ 0.77 kg/cm$^2$, 9 ~ 11 psi) |
| | rear | 91 ~ 105 kPa (0.91 ~ 1.05 kg/cm$^2$, 13 ~ 15 psi) |
| *Load is total weight of cargo, rider, passenger, accessories, and tongue | | |
| **Brake** | | |
| Front brake | type | Dual disc brake |
| | operation | Foot operation |
| Rear brake | type | Dual disc brake |
| | operation | Foot operation |
| Parking brake | type | Single disc brake |
| | operation | Right hand operation |
| **Suspension** | | |
| Front suspension | | Double wishbone |
| Rear suspension | | Double wishbone |
| **Shock absorber** | | |
| Front shock absorber | | Coil spring/oil damper *1, *2<br>Coil spring/gas-oil damper *3, *4 |
| Rear shock absorber | | Coil spring/oil damper *1, *2<br>Coil spring/gas-oil damper *3, *4 |
| **Wheel travel** | | |
| Front wheel travel | | 180 mm (7.1 in) |
| Rear wheel travel | | 180 mm (7.1 in) |
| **Electrical system** | | |
| Ignition system | | Transistorized coil ignition (digital) |
| Charging system | | AC magneto |
| **Battery** | | |
| Model | | U1L-11 |
| Battery capacity | | 12 V 28.0 Ah |
| Manufacturer | | GS YUASA |
| **Bulb type** | | Krypton bulb |

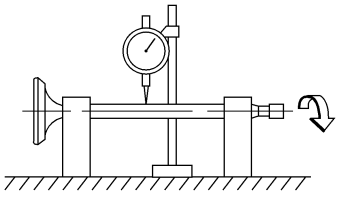
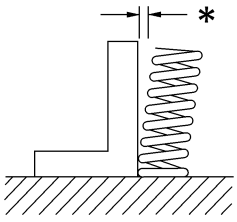
| Item | Standard |
|---|---|
| **Bulb voltage/wattage × quantity** | |
| Headlight | 12 V 30.0 W/30.0 W × 2 |
| Tail/brake light | 12 V 5.0/21.0 W × 2 |
| Indicator light | |
| Neutral indicator light | LED |
| Reverse indicator light | LED |
| Coolant temperature warning light | LED |
| Engine trouble warning light | LED |
| Park indicator light | LED |
| On-command four-wheel drive/differential gear lock indicator | LCD |
| High-range indicator light | LED |
| Low-range indicator light | LED |
| Differential gear lock indicator light | LED |

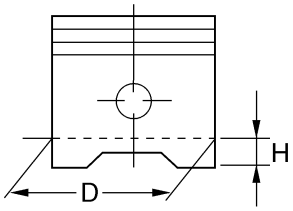
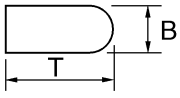
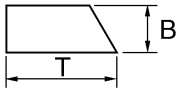
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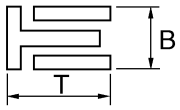
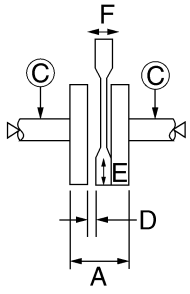
# ENGINE SPECIFICATIONS

| Item | Standard | Limit |
|---|---|---|
| **Cylinder head** | | |
| Maximum warpage ∗  | ---- | 0.03 mm (0.0012 in) |
| **Cylinder** | | |
| Bore | 102.000 ~ 102.010 mm (4.0157 ~ 4.0161 in) | 102.080 mm (4.0189 in) |
| Measuring point ∗  | 50.0 mm (1.97 in) | ---- |
| Maximum taper | | 0.05 mm (0.002 in) |
| Out of round | | 0.05 mm (0.002 in) |
| **Camshaft** | | |
| Drive system | Chain drive | ---- |
| Camshaft lobe dimensions  | | |
| Intake measurement “A” | 42.481 ~ 42.581 mm (1.6725 ~ 1.6764 in) | 42.381 mm (1.6685 in) |
| “B” | 36.950 ~ 37.050 mm (1.4547 ~ 1.4587 in) | 36.850 mm (1.4508 in) |
| Exhaust measurement “A” | 43.129 ~ 43.229 mm (1.6980 ~ 1.7019 in) | 43.029 mm (1.6941 in) |
| “B” | 36.982 ~ 37.082 mm (1.4560 ~ 1.4599 in) | 36.882 mm (1.4520 in) |
| Maximum camshaft runout  | ---- | 0.015 mm (0.0006 in) |

| Item | Standard | Limit |
|---|---|---|
| **Timing chain** | | |
| Model/number of links | 98XRH2010/126 | ---- |
| Tensioning system | Automatic | ---- |
| **Rocker arm/rocker arm shaft** | | |
| Rocker arm inside diameter | 12.000 ~ 12.018 mm (0.4724 ~ 0.4731 in) | ---- |
| Shaft outside diameter | 11.981 ~ 11.991 mm (0.4717 ~ 0.4721 in) | ---- |
| Rocker-arm-to-rocker-arm-shaft clearance | 0.009 ~ 0.037 mm (0.0004 ~ 0.0015 in) | ---- |
| **Valve, valve seat, valve guide** | | |
| Valve clearance–intake (cold) | 0.09 ~ 0.13 mm (0.0035 ~ 0.0051 in) | ---- |
| Valve clearance–exhaust (cold) | 0.16 ~ 0.20 mm (0.0063 ~ 0.0079 in) | ---- |
| Valve dimensions | | |
|  Head Diameter |  Face Width |  Seat Width |  Margin Thickness |
| Valve head diameter “A” | | |
| Intake | 37.90 ~ 38.10 mm (1.4921 ~ 1.5000 in) | ---- |
| Exhaust | 31.90 ~ 32.10 mm (1.2559 ~ 1.2638 in) | ---- |
| Valve face width “B” | | |
| Intake | 2.26 mm (0.0890 in) | ---- |
| Exhaust | 2.26 mm (0.0890 in) | ---- |
| Valve seat width “C” | | |
| Intake | 1.00 ~ 1.20 mm (0.0394 ~ 0.0472 in) | 1.60 mm (0.0630 in) |
| Exhaust | 1.00 ~ 1.20 mm (0.0394 ~ 0.0472 in) | 1.60 mm (0.0630 in) |
| Valve margin thickness “D” | | |
| Intake | 0.80 ~ 1.20 mm (0.0315 ~ 0.0472 in) | ---- |
| Exhaust | 0.80 ~ 1.20 mm (0.0315 ~ 0.0472 in) | ---- |
| Valve stem diameter | | |
| Intake | 5.975 ~ 5.990 mm (0.2352 ~ 0.2358 in) | 5.945 mm (0.2341 in) |
| Exhaust | 5.960 ~ 5.975 mm (0.2346 ~ 0.2352 in) | 5.930 mm (0.2335 in) |
| Valve guide inside diameter | | |
| Intake | 6.000 ~ 6.012 mm (0.2362 ~ 0.2367 in) | 6.050 mm (0.2382 in) |
| Exhaust | 6.000 ~ 6.012 mm (0.2362 ~ 0.2367 in) | 6.050 mm (0.2382 in) |

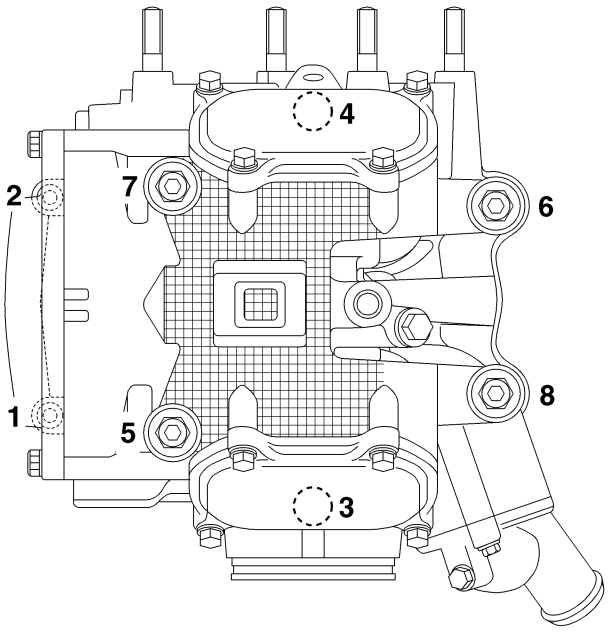
| Item | Standard | Limit |
|---|---|---|
| Valve-stem-to-valve-guide clearance | | |
| Intake | 0.010 ~ 0.037 mm (0.0004 ~ 0.0015 in) | 0.080 mm (0.0031 in) |
| Exhaust | 0.025 ~ 0.052 mm (0.0010 ~ 0.0020 in) | 0.100 mm (0.0039 in) |
| Valve stem runout | ---- | 0.040 mm (0.0016 in) |
|  | | |
| **Valve spring** | | |
| Free length | | |
| Intake | 40.38 mm (1.59 in) | 38.36 mm (1.51 in) |
| Exhaust | 40.38 mm (1.59 in) | 38.36 mm (1.51 in) |
| Installed length (valve closed) | | |
| Intake | 35.00 mm (1.38 in) | ---- |
| Exhaust | 35.00 mm (1.38 in) | ---- |
| Compressed spring force (installed) | | |
| Intake | 171 ~ 197 N (17.44 ~ 20.09 kgf, 38.44 ~ 44.29 lb) | ---- |
| Exhaust | 171 ~ 197 N (17.44 ~ 20.09 kgf, 38.44 ~ 44.29 lb) | ---- |
| Spring tilt * | | |
|  | | |
| Intake | ---- | 2.5°/1.80 mm (2.5°/0.071 in) |
| Exhaust | ---- | 2.5°/1.80 mm (2.5°/0.071 in) |
| Winding direction (top view) | | |
| Intake | Clockwise | ---- |
| Exhaust | Clockwise | ---- |

| Item | Standard | Limit |
|---|---|---|
| **Piston** | | |
| Piston-to-cylinder clearance | 0.030 ~ 0.055 mm (0.0012 ~ 0.0022 in) | 0.13 mm (0.0051 in) |
|  | | |
| Diameter “D” | 101.955 ~ 101.970 mm (4.0140 ~ 4.0146 in) | ---- |
| Height “H” | 10.0 mm (0.39 in) | ---- |
| Offset | 0.50 mm (0.0197 in) | ---- |
| Offset direction | Intake side | ---- |
| Piston pin bore inside diameter | 23.004 ~ 23.015 mm (0.9057 ~ 0.9061 in) | 23.045 mm (0.9073 in) |
| Piston pin outside diameter | 22.991 ~ 23.000 mm (0.9052 ~ 0.9055 in) | 22.971 mm (0.9044 in) |
| Piston-pin-to-piston-pin-bore clearance | 0.004 ~ 0.024 mm (0.0002 ~ 0.0009 in) | 0.074 mm (0.0029 in) |
| **Piston rings** | | |
| Top ring | | |
|  | | |
| Ring type | Barrel | ---- |
| Dimensions (B × T) | 1.20 × 3.80 mm (0.05 × 0.15 in) | ---- |
| End gap (installed) | 0.20 ~ 0.35 mm (0.008 ~ 0.014 in) | 0.60 mm (0.024 in) |
| Ring side clearance | 0.030 ~ 0.070 mm (0.0012 ~ 0.0028 in) | 0.12 mm (0.0047 in) |
| 2nd ring | | |
|  | | |
| Ring type | Taper | ---- |
| Dimensions (B × T) | 1.20 × 4.00 mm (0.05 × 0.16 in) | ---- |
| End gap (installed) | 0.75 ~ 0.90 mm (0.030 ~ 0.035 in) | 1.25 mm (0.049 in) |
| Ring side clearance | 0.030 ~ 0.070 mm (0.0012 ~ 0.0028 in) | 0.13 mm (0.0051 in) |

| Item | Standard | Limit |
|---|---|---|
| Oil ring   | | |
| Dimensions (B × T) | 2.50 × 2.80 mm (0.10 × 0.11 in) | ---- |
| End gap (installed) | 0.20 ~ 0.70 mm (0.008 ~ 0.028 in) | ---- |
| Ring side clearance | 0.060 ~ 0.150 mm (0.0024 ~ 0.0059 in) | ---- |
| **Crankshaft**   | | |
| Crank width “A” | 74.95 ~ 75.00 mm (2.951 ~ 2.953 in) | ---- |
| Maximum runout “C” | ---- | 0.030 mm (0.0012 in) |
| Big end side clearance “D” | 0.350 ~ 0.650 mm (0.0138 ~ 0.0256 in) | 1.0 mm (0.04 in) |
| Big end radial clearance “E” | 0.010 ~ 0.025 mm (0.0004 ~ 0.0010 in) | ---- |
| Small end free play “F” | 0.16 ~ 0.40 mm (0.0063 ~ 0.0157 in) | ---- |
| **Balancer** | | |
| Balancer drive method | Gear | ---- |
| **Automatic centrifugal clutch** | | |
| Clutch shoe thickness | 1.5 mm (0.06 in) | 1.0 mm (0.04 in) |
| Clutch-in revolution | 1,850 ~ 2,250 r/min | ---- |
| Clutch-stall revolution | 3,500 ~ 4,100 r/min | ---- |
| **V-belt** | | |
| V-belt width | 33.3 mm (1.31 in) | 30.0 mm (1.18 in) |
| **Transmission** | | |
| Maximum main axle runout | ---- | 0.06 mm (0.0024 in) |
| Maximum drive axle runout | ---- | 0.06 mm (0.0024 in) |
| **Shifting mechanism** | | |
| Shift mechanism type | Shift drum and guide bar | ---- |
| **Decompression device** | | |
| Device type | Auto decomp | ---- |
| **Air filter oil grade** | Foam air filter oil or equivalent oil | ---- |

| Item | Standard | Limit |
|---|---|---|
| **Throttle body** | | |
| Model/manufacturer × quantity | 41EHS/MIKUNI × 1 | ---- |
| Engine idle speed | 1,550 ~ 1,650 r/min | ---- |
| Intake vacuum | 40.0 kPa (300 mmHg, 11.8 inHg) | ---- |
| Water temperature | 80 ~ 90 °C | ---- |
| Oil temperature | 60 ~ 70 °C | ---- |
| **Fuel pump** | | |
| Pump type | Electrical | ---- |
| Model/manufacturer | 5B4/AISAN | ---- |
| **Oil filter type** | Cartridge (paper) | ---- |
| **Oil pump** | | |
| Oil pump type | Trochoid | ---- |
| Inner-rotor-to-outer-rotor-tip clearance | Less than 0.12 mm (0.0047 in) | 0.20 mm (0.0079 in) |
| Outer-rotor-to-oil-pump-housing clearance | 0.090 ~ 0.170 mm (0.0035 ~ 0.0067 in) | 0.24 mm (0.0094 in) |
| Oil-pump-housing-to-inner-and-outer-rotor clearance | 0.03 ~ 0.10 mm (0.0012 ~ 0.0039 in) | 0.17 mm (0.0067 in) |
| Oil pressure (hot) | 50.0 kPa at 1,600 r/min (0.50 kg/cm$^2$ at 1,600 r/min, 7.1 psi at 1,600 r/min) | ---- |
| Pressure check location | Cylinder head | ---- |
| **Cooling system** | | |
| Radiator core | | |
| Width | 400.0 mm (15.75 in) | ---- |
| Height | 266.0 mm (10.47 in) | ---- |
| Depth | 26.0 mm (1.02 in) | ---- |
| Radiator cap opening pressure | 93.3 ~ 122.7 kPa (0.933 ~ 1.227 kg/cm$^2$, 13.27 ~ 17.45 psi) | ---- |
| Coolant reservoir capacity | | |
| Up to the maximum level mark | 0.32 L (0.28 Imp qt, 0.34 US qt) | ---- |
| From low to full level | 0.21 L (0.18 Imp qt, 0.22 US qt) | ---- |
| **Water pump** | | |
| Water pump type | Single-suction centrifugal pump | ---- |
| Reduction ratio | 32/31 (1.032) | ---- |
| **Shaft drive** | | |
| Middle gear backlash | 0.10 ~ 0.30 mm (0.0039 ~ 0.0118 in) | ---- |
| Final gear backlash | 0.10 ~ 0.30 mm (0.0039 ~ 0.0118 in) | ---- |
| Differential gear backlash | 0.05 ~ 0.25 mm (0.0020 ~ 0.0098 in) | ---- |

**Cylinder head tightening sequence**



EBS01003

# CHASSIS SPECIFICATIONS

*1 For models equipped with panel wheels and oil damper shock absorbers
*2 For models equipped with cast wheels and oil damper shock absorbers
*3 For models equipped with cast wheels and gas-oil damper shock absorbers
*4 For models equipped with cast wheels, gas-oil damper shock absorbers, and overfenders

| Item | Standard | Limit |
|---|---|---|
| **Steering** | | |
| Steering bearing type | Ball and race bearing | ---- |
| **Front suspension** | | |
| Shock absorber stroke | 108.0 mm (4.25 in) *1, *2 | ---- |
| | 110.0 mm (4.33 in) *3, *4 | ---- |
| Spring free length | 313.0 mm (12.32 in) *1, *2 | ---- |
| | 336.6 mm (13.25 in) *3, *4 | ---- |
| Installed length | 247.9 mm (9.76 in) *1, *2 | ---- |
| | 260.5 mm (10.26 in) *3, *4 | ---- |
| Spring rate (K1) | 19.40 N/mm (1.98 kgf/mm, 110.77 lb/in) | ---- |
| Spring stroke (K1) | 0.0 ~ 108.0 mm (0.00 ~ 4.25 in) *1, *2 | ---- |
| | 0.0 ~ 110.0 mm (0.00 ~ 4.33 in) *3, *4 | ---- |
| Optional spring available | No | ---- |
| **Rear suspension** | | |
| Rear shock absorber assembly | 81.0 mm (3.19 in) *1, *2 | ---- |
| | 85.0 mm (3.35 in) *3, *4 | ---- |
| Spring free length | 328.0 mm (12.91 in) *1, *2 | ---- |
| | 286.3 mm (11.27 in) *3, *4 | ---- |
| Installed length | 273.2 mm (10.76 in) *1, *2 | ---- |
| | 235.0 mm (9.25 in) *3, *4 | ---- |
| Spring rate (K1) | 44.10 N/mm (4.50 kgf/mm, 251.81 lb/in) | ---- |
| Spring rate (K2) | 117.70 N/mm (12.00 kgf/mm, 672.07 lb/in) *1, *2 | ---- |
| Spring stroke (K1) | 0.0 ~ 60.0 mm (0.00 ~ 2.36 in) *1, *2 | ---- |
| | 0.0 ~ 85.0 mm (0.00 ~ 3.35 in) *3, *4 | ---- |
| Spring stroke (K2) | 60.0 ~ 81.0 mm (2.36 ~ 3.19 in) *1, *2 | ---- |
| Optional spring available | No | ---- |
| **Front wheel** | | |
| Wheel type | Panel wheel *1 | ---- |
| | Cast wheel *2, *3, *4 | ---- |
| Rim size | 12 × 6.0 AT | ---- |
| Rim material | Steel *1 | ---- |
| | Aluminum *2, *3, *4 | ---- |
| Max. radial wheel runout | ---- | 2.0 mm (0.08 in) |
| Max. lateral wheel runout | ---- | 2.0 mm (0.08 in) |

| Item | Standard | Limit |
|---|---|---|
| **Rear wheel** | | |
| Wheel type | Panel wheel *1 | ---- |
| | Cast wheel *2, *3, *4 | ---- |
| Rim size | 12 × 7.5 AT | ---- |
| Rim material | Steel *1 | ---- |
| | Aluminum *2, *3, *4 | ---- |
| Max. radial wheel runout | ---- | 2.0 mm (0.08 in) |
| Max. lateral wheel runout | ---- | 2.0 mm (0.08 in) |
| **Front disc brake** | | |
| Type | Dual | ---- |
| Disc outside diameter × thickness | 200.0 × 3.5 mm (7.87 × 0.14 in) | ---- |
| Brake disc minimum thickness | 3.0 mm (0.12 in) | ---- |
| Brake disc maximum deflection | 0.1 mm (0.004 in) | ---- |
| Pad thickness inner | 5.2 mm (0.20 in) | 1.5 mm (0.06 in) |
| Pad thickness outer | 5.2 mm (0.20 in) | 1.5 mm (0.06 in) |
| Master cylinder inside diameter | 17.46 mm (0.69 in) | ---- |
| Caliper cylinder inside diameter | 27.00 mm (1.06 in) | ---- |
| Recommended fluid | DOT 4 | ---- |
| **Rear disc brake** | | |
| Type | Dual | ---- |
| Disc outside diameter × thickness | 184.6 × 3.5 mm (7.27 × 0.14 in) | ---- |
| Brake disc minimum thickness | 3.0 mm (0.12 in) | ---- |
| Brake disc maximum deflection | 0.1 mm (0.004 in) | ---- |
| Pad thickness inner | 5.2 mm (0.20 in) | 1.5 mm (0.06 in) |
| Pad thickness outer | 5.2 mm (0.20 in) | 1.5 mm (0.06 in) |
| Caliper cylinder inside diameter | 25.40 mm (1.00 in) | ---- |
| Recommended fluid | DOT 4 | ---- |

| Item | Standard | Limit |
|---|---|---|
| **Parking brake** | | |
| Type | Disc | ---- |
| Disc outside diameter × thickness | 165.0 × 3.5 mm (6.50 × 0.14 in) | ---- |
| Brake disc minimum thickness | 3.0 mm (0.12 in) | ---- |
| Brake disc maximum deflection | 0.1 mm (0.004 in) | ---- |
| Brake pad lining thickness-inner | 3.2 mm (0.13 in) | 1.0 mm (0.04 in) |
| Brake pad lining thickness-outer | 3.2 mm (0.13 in) | 1.0 mm (0.04 in) |
| **Brake lever and brake pedal** | | |
| Accelerator pedal free play | 0 mm (0 in) | ---- |
| Brake pedal free play | 0 mm (0 in) | ---- |
| Parking brake cable free play | The maximum free play is equal to one click of the parking brake lever. | ---- |

EBS01004

# ELECTRICAL SPECIFICATIONS

| Item | Standard | Limit |
|---|---|---|
| **System voltage** | 12 V | ---- |
| **Ignition system** | | |
| Ignition timing (B.T.D.C.) | 12°/1,400 r/min | ---- |
| Advancer type | Digital | ---- |
| **Transistorized coil ignition** | | |
| Crankshaft position sensor resistance/color | 459 ~ 561 Ω at 20 °C (68 °F)/ gray/black | ---- |
| **ECU (engine control unit)** | | |
| Model/manufacturer | F8T83871/MITSUBISHI | ---- |
| **Ignition coil** | | |
| Model/manufacturer | 2JN/YAMAHA | ---- |
| Minimum ignition spark gap | 6.0 mm (0.24 in) | ---- |
| Primary coil resistance | 2.16 ~ 2.64 Ω at 20 °C (68 °F) | ---- |
| Secondary coil resistance | 8.64 ~ 12.96 kΩ at 20 °C (68 °F) | ---- |
| **Spark plug cap** | | |
| Material | Resin | ---- |
| Resistance | 10.0 kΩ | ---- |
| **AC magneto** | | |
| Model/manufacturer | F004T39372/MITSUBISHI | ---- |
| Standard output | 14.0 V 33.5 A at 5,000 r/min | ---- |
| Stator coil resistance/color | 0.099 ~ 0.121 Ω at 20 °C (68 °F)/ white-white | ---- |
| **Rectifier/regulator** | | |
| Type | Semiconductor-short-circuit | ---- |
| Model/manufacturer | FH012AA/SHINDENGEN | ---- |
| No load regulated voltage (DC) | 14.2 ~ 14.8 V | ---- |
| Rectifier capacity | 50.0 A | ---- |
| Withstand voltage | 40.0 V | ---- |
| **Electric starting system** | | |
| Type | Constant mesh | ---- |
| **Starter motor** | | |
| Model/manufacturer | SM-13/MITSUBA | ---- |
| Power output | 0.80 kW | ---- |
| Armature coil resistance | 0.0250 ~ 0.0350 Ω at 20 °C (68 °F) | ---- |
| Brush overall length | 12.5 mm (0.49 in) | 5.00 mm (0.20 in) |
| Spring force | 7.65 ~ 10.01 N (780 ~ 1,021 gf, 27.54 ~ 36.03 oz) | ---- |
| Commutator diameter | 28.0 mm (1.10 in) | 27.0 mm (1.06 in) |
| Mica undercut (depth) | 0.70 mm (0.03 in) | ---- |

| Item | Standard | Limit |
|---|---|---|
| **Starter relay** | | |
| Model/manufacturer | RC19-080A/MITSUBA | ---- |
| Amperage rating | 180.0 A | ---- |
| Coil winding resistance | 4.18 ~ 4.62 Ω at 20 °C (68 °F) | ---- |
| **Fuel sender** | | |
| Sender unit resistance (full) | 19.00 ~ 21.00 Ω | ---- |
| Sender unit resistance (empty) | 137.00 ~ 143.00 Ω | ---- |
| **Fuel injection system relay** | | |
| Model/manufacturer | ACM33211/MATSUSHITA | ---- |
| Coil resistance | 96.0 Ω | ---- |
| **Radiator fan motor relay** | | |
| Model/manufacturer | ACM33211/MATSUSHITA | ---- |
| Coil resistance | 96.0 Ω | ---- |
| **Load control relay** | | |
| Model/manufacturer | ACM33211/MATSUSHITA | ---- |
| Coil resistance | 96.0 Ω | ---- |
| **Four-wheel-drive motor relay 3** | | |
| Model/manufacturer | ACM33211/MATSUSHITA | ---- |
| Coil resistance | 96.0 Ω | ---- |
| **Headlight relay** | | |
| Model/manufacturer | G8HN-1C4T-DJ/OMRON | ---- |
| Coil resistance | 105.0 Ω | ---- |
| **Four-wheel-drive motor relay 1** | | |
| Model/manufacturer | G8HN-1C4T-DJ/OMRON | ---- |
| Coil resistance | 105.0 Ω | ---- |
| **Four-wheel-drive motor relay 2** | | |
| Model/manufacturer | G8HN-1C4T-DJ/OMRON | ---- |
| Coil resistance | 105.0 Ω | ---- |
| **Circuit breaker** | | |
| Circuit breaker type | Fuse | ---- |
| Fuses | | |
| Main fuse | 40.0 A | ---- |
| Headlight fuse | 15.0 A | ---- |
| Signaling system fuse | 10.0 A | ---- |
| Ignition fuse | 10.0 A | ---- |
| Auxiliary DC jack fuse | 10.0 A | ---- |
| Fuel injection system fuse | 10.0 A | ---- |
| Four-wheel-drive motor fuse | 10.0 A | ---- |
| Backup fuse | 10.0 A | ---- |
| Radiator fan motor fuse | 25.0 A | ---- |
| Spare fuse | 25.0 A | ---- |
| | 15.0 A | ---- |
| | 10.0 A | ---- |

EBS01005

# TIGHTENING TORQUES

## ENGINE TIGHTENING TORQUES

| Item | Part name | Thread size | Q'ty | Tightening torque | | | Remarks |
|---|---|---|---|---|---|---|---|
| | | | | Nm | m · kg | ft · lb | |
| Cylinder head (exhaust pipe) | Stud bolt | M8 | 4 | 15 | 1.5 | 11 | |
| Cylinder head | Bolt | M9 | 4 | 35 | 3.5 | 25 | M |
| Cylinder head | Bolt | M9 | 2 | 38 | 3.8 | 27 | E |
| Cylinder head | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Spark plug | — | M10 | 1 | 13 | 1.3 | 9.4 | |
| Oil check bolt | Bolt | M8 | 1 | 10 | 1.0 | 7.2 | |
| Cylinder | Bolt | M10 | 4 | 50 | 5.0 | 36 | See TIP. E |
| AC magneto rotor | Nut | M16 | 1 | 60 | 6.0 | 43 | E |
| Balancer driven gear | Nut | M18 | 1 | 80 | 8.0 | 58 | Use a lock washer. E |
| Thermostat cover | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Cylinder head air bleed bolt | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Valve adjusting screw | Nut | M6 | 4 | 14 | 1.4 | 10 | |
| Decompression assembly | Bolt | M7 | 2 | 20 | 2.0 | 14 | |
| Timing chain guide (intake side) | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | LT |
| Timing chain tensioner cap | Bolt | M16 | 1 | 20 | 2.0 | 14 | |
| Timing chain tensioner | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Bearing retainer (camshaft) | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | LT |
| Camshaft sprocket cover | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Tappet cover | Bolt | M6 | 8 | 10 | 1.0 | 7.2 | |
| Camshaft sprocket | Bolt | M7 | 2 | 20 | 2.0 | 14 | |
| Crankcase | Bolt | M8 | 3 | 26 | 2.6 | 19 | |
| | Bolt | M6 | 4 | 10 | 1.0 | 7.2 | |
| | Bolt | M6 | 8 | 10 | 1.0 | 7.2 | |
| Engine oil drain bolt | Bolt | M14 | 1 | 30 | 3.0 | 22 | |
| Oil filter cartridge | — | M20 | 1 | 17 | 1.7 | 12 | |
| Oil filter cartridge union bolt | Union bolt | M20 | 1 | 68 | 6.8 | 49 | E |
| Oil pipe (dipstick) | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Oil delivery pipe 1 | Union bolt | M14 | 2 | 35 | 3.5 | 25 | |
| Oil delivery pipe 1 | Union bolt | M10 | 1 | 20 | 2.0 | 14 | |
| Oil delivery pipe 1 | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Oil delivery pipe 2 | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Oil delivery pipe 3 | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Oil pump | Bolt | M6 | 3 | 10 | 1.0 | 7.2 | |
| Oil pump driven gear | Nut | M10 | 1 | 22 | 2.2 | 16 | |
| Timing chain guide | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | LT |
| V-belt cooling duct 1 | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Air filter case | Bolt | M6 | 4 | 7 | 0.7 | 5.1 | |

| Item | Part name | Thread size | Q'ty | Tightening torque | | | Remarks |
|---|---|---|---|---|---|---|---|
| | | | | Nm | m · kg | ft · lb | |
| Air intake duct | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Muffler | Bolt | M8 | 2 | 20 | 2.0 | 14 | |
| Muffler bracket | Bolt | M8 | 4 | 30 | 3.0 | 22 | |
| Exhaust pipe | Nut | M8 | 4 | 20 | 2.0 | 14 | |
| Tailpipe | Bolt | M6 | 3 | 10 | 1.0 | 7.2 | |
| Radiator inlet pipe holder | Bolt | M6 | 1 | 7 | 0.7 | 5.1 | |
| Radiator outlet pipe holder | Bolt | M6 | 1 | 7 | 0.7 | 5.1 | |
| Oil cooler inlet pipe 1/oil cooler outlet pipe 1 holder | Bolt | M6 | 1 | 7 | 0.7 | 5.1 | |
| Oil cooler inlet pipe 2/oil cooler outlet pipe 2 holder | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Oil cooler inlet pipe 1/oil cooler outlet pipe 1 and oil cooler | Bolt | M17 | 2 | 21 | 2.1 | 15 | |
| Oil cooler | Nut | M6 | 4 | 7 | 0.7 | 5.1 | |
| Radiator | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Coolant reservoir | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Water pump housing | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Coolant drain bolt | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Water pump air bleed bolt | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Water pump outlet pipe | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Water jacket joint | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Crankshaft end accessing screw | Bolt | M36 | 1 | 10 | 1.0 | 7.2 | |
| Timing mark accessing screw | Bolt | M14 | 1 | 6 | 0.6 | 4.3 | |
| Drive belt cover | Bolt | M6 | 12 | 10 | 1.0 | 7.2 | |
| Drive belt case | Bolt | M6 | 8 | 10 | 1.0 | 7.2 | |
| V-belt cooling duct 1 bracket | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Bearing housing (primary sheave assembly) | Bolt | M6 | 4 | 10 | 1.0 | 7.2 | |
| Bearing retainer (bearing housing) | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| AC magneto cover | Bolt | M6 | 11 | 10 | 1.0 | 7.2 | |
| AC magneto/crankshaft position sensor lead holder | Bolt | M5 | 2 | 7 | 0.7 | 5.1 |  |
| Starter one-way clutch | Screw | M8 | 3 | 30 | 3.0 | 22 |  |
| Clutch housing assembly | Bolt | M6 | 9 | 10 | 1.0 | 7.2 | |
| Clutch carrier assembly | Nut | M22 | 1 | 190 | 19.0 | 140 | Stake.  |
| Middle drive pinion gear nut | Nut | M22 | 1 | 190 | 19.0 | 140 | Stake. |
| Middle driven shaft bearing housing | Bolt | M8 | 4 | 32 | 3.2 | 23 |  |
| Middle drive shaft bearing retainer | Screw | M8 | 4 | 29 | 2.9 | 21 | Stake.  |
| Middle driven shaft bearing retainer | Nut | M55 | 1 | 80 | 8.0 | 58 | Left-hand thread  |

| Item | Part name | Thread size | Q'ty | Tightening torque | | | Remarks |
|---|---|---|---|---|---|---|---|
| | | | | Nm | m · kg | ft · lb | |
| Middle driven pinion gear bearing retainer | Nut | M60 | 1 | 130 | 13.0 | 94 | Left-hand thread |
| Universal joint yoke nut (middle gear side) | Nut | M16 | 1 | 150 | 15.0 | 110 | |
| Middle driven pinion gear bearing housing | Bolt | M8 | 4 | 25 | 2.5 | 18 | |
| Drive shaft coupling gear nut (middle gear side) | Nut | M16 | 1 | 190 | 19.0 | 140 | |
| Primary sheave assembly | Nut | M16 | 1 | 140 | 14.0 | 100 | |
| Secondary sheave spring retainer | Nut | M36 | 1 | 90 | 9.0 | 65 | |
| Secondary sheave assembly | Nut | M16 | 1 | 100 | 10.0 | 72 | |
| Shift lever cover | Bolt | M6 | 4 | 10 | 1.0 | 7.2 | |
| Shift lever 2 assembly | Bolt | M6 | 1 | 14 | 1.4 | 10 | |
| Shift drum stopper | Bolt | M14 | 1 | 18 | 1.8 | 13 | |
| Crankcase plug bolt | Bolt | M14 | 1 | 18 | 1.8 | 13 | |
| Select lever unit | Bolt | M8 | 3 | 15 | 1.5 | 11 | |
| Shift arm | Bolt | M6 | 1 | 14 | 1.4 | 10 | |
| Shift rod locknut (select lever unit side) | Nut | M8 | 1 | 15 | 1.5 | 11 | Left-hand thread |
| Shift rod locknut (shift arm side) | Nut | M8 | 1 | 15 | 1.5 | 11 | |
| Stator coil assembly | Bolt | M6 | 3 | 7 | 0.7 | 5.1 | |
| Crankshaft position sensor | Bolt | M5 | 2 | 7 | 0.7 | 5.1 | |
| Coolant temperature sensor | — | M12 | 1 | 18 | 1.8 | 13 | |
| Gear position switch | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Reverse switch | — | M10 | 1 | 17 | 1.7 | 12 | |
| Speed sensor | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |

**TIP:**
Temporarily tighten the cylinder bolts to 15 Nm (1.5 m · kg, 11 ft · lb) and then tighten them to 50 Nm (5.0 m · kg, 36 ft · lb).

EBS01006

## CHASSIS TIGHTENING TORQUES

| Part to be tightened | Thread size | Tightening torque | | | Remarks |
|---|---|---|---|---|---|
| | | Nm | m · kg | ft · lb | |
| Engine and front rubber damper | M10 | 42 | 4.2 | 30 | |
| Engine and front rubber damper | M6 | 10 | 1.0 | 7.2 | |
| Engine and rear rubber damper | M10 | 42 | 4.2 | 30 | |
| Engine and rear rubber damper | M6 | 10 | 1.0 | 7.2 | |
| Rubber damper and frame | M10 | 42 | 4.2 | 30 | |
| Front skid plate and frame | M6 | 7 | 0.7 | 5.1 | |
| Rear skid plate and frame | M6 | 7 | 0.7 | 5.1 | |
| Seat support and frame | M8 | 16 | 1.6 | 11 | |
| Seat support bracket and seat support | M10 | 32 | 3.2 | 23 | |
| Heat protector and frame | M6 | 8 | 0.8 | 5.8 | |
| Upper instrument panel and frame | M6 | 7 | 0.7 | 5.1 | |
| Passenger seat and passenger handhold bracket | M6 | 8 | 0.8 | 5.8 | |
| Passenger handhold strap and passenger handhold bracket | M6 | 7 | 0.7 | 5.1 | |
| Fuel tank holder and frame | M8 | 30 | 3.0 | 22 | |
| Passenger seat damper plate and frame | M8 | 16 | 1.6 | 11 | |
| Side door bracket and side door | M6 | 7 | 0.7 | 5.1 | |
| Frame and hinge | M8 | 23 | 2.3 | 17 | |
| Frame and latch | M6 | 8 | 0.8 | 5.8 | |
| Handle latch and side door | M5 | 5 | 0.5 | 3.6 | |
| Fuel tank and fuel pump | M6 | 8 | 0.8 | 5.8 | |
| Front bumper and frame | M12 | 59 | 5.9 | 43 | |
| | M10 | 32 | 3.2 | 23 | |
| Cargo bed release lever | M8 | 16 | 1.6 | 11 | |
| | M6 | 7 | 0.7 | 5.1 | |
| Gas spring assembly and cargo bed assembly | M8 | 16 | 1.6 | 11 | |
| Gas spring assembly and frame | M8 | 16 | 1.6 | 11 | |
| Gas spring assembly bracket and cargo bed assembly | M6 | 7 | 0.7 | 5.1 | |
| Cargo hook and cargo bed | M6 | 7 | 0.7 | 5.1 | |
| Tailgate cable and cargo bed | M6 | 7 | 0.7 | 5.1 | |
| Tailgate cable and tailgate | M6 | 7 | 0.7 | 5.1 | |
| Cargo bed panel and cargo bed assembly | M6 | 7 | 0.7 | 5.1 | |
| Cargo bed latch and tailgate | M6 | 9 | 0.9 | 6.5 | |
| Tail/brake light and cargo bed assembly | M6 | 7 | 0.7 | 5.1 | |
| Side frame (enclosure) and frame | M10 | 65 | 6.5 | 47 | |
| | M8 | 22 | 2.2 | 16 | |
| Lower support frame (enclosure) and frame | M10 | 64 | 6.4 | 46 | |
| Upper support frame (enclosure) and lower support frame (enclosure) | M8 | 22 | 2.2 | 16 | |
| Support frame (enclosure) and side frame (enclosure) | M10 | 65 | 6.5 | 47 | |

| Part to be tightened | Thread size | Tightening torque | | | Remarks |
|---|---|---|---|---|---|
| | | Nm | m · kg | ft · lb | |
| Top frame (enclosure) and side frame (enclosure) | M10 | 65 | 6.5 | 47 | |
| Headrest and upper support frame | M6 | 7 | 0.7 | 5.1 | |
| Seat belt and frame | M10 | 59 | 5.9 | 43 | |
| Seat belt and side frame (enclosure) | M10 | 59 | 5.9 | 43 | |
| Buckle and lower support frame | M10 | 59 | 5.9 | 43 | |
| Rectifier/regulator and frame | M6 | 7 | 0.7 | 5.1 | |
| Engine ground lead and crankcase | M6 | 10 | 1.0 | 7.2 | |
| Starter motor lead and starter motor | M5 | 5 | 0.5 | 3.6 | |
| ECU (engine control unit) and electrical components tray | M6 | 7 | 0.7 | 5.1 | |
| Ignition coil and frame | M6 | 7 | 0.7 | 5.1 | |
| Steering assembly and steering joint | M8 | 22 | 2.2 | 16 | |
| Steering shaft and steering joint | M8 | 22 | 2.2 | 16 | |
| Steering assembly and frame | M10 | 48 | 4.8 | 35 | |
| Steering shaft and frame | M8 | 21 | 2.1 | 15 | |
| Steering wheel and steering shaft assembly | M12 | 35 | 3.5 | 25 | |
| Tie-rod locknut | M12 | 40 | 4.0 | 29 | |
| Front upper arm and frame | M10 | 45 | 4.5 | 32 |  |
| Front lower arm and frame | M10 | 45 | 4.5 | 32 |  |
| Front shock absorber and frame | M10 | 45 | 4.5 | 32 | |
| Front shock absorber and front upper arm | M10 | 45 | 4.5 | 32 | |
| Steering knuckle and tie-rod | M12 | 39 | 3.9 | 28 | |
| Steering knuckle and front upper arm | M12 | 30 | 3.0 | 22 | |
| Steering knuckle and front lower arm | M12 | 30 | 3.0 | 22 | |
| Steering knuckle and brake disc guard | M6 | 7 | 0.7 | 5.1 | |
| Front arm protector and front lower arm | M6 | 7 | 0.7 | 5.1 | |
| Front wheel and front wheel hub | M10 | 55 | 5.5 | 40 | |
| Front wheel hub and constant velocity joint | M20 | 350 | 35.0 | 250 | Stake. |
| Front brake caliper and steering knuckle | M10 | 48 | 4.8 | 35 | |
| Front brake disc and front wheel hub | M8 | 30 | 3.0 | 22 |  |
| Front brake hose union bolt | M10 | 27 | 2.7 | 19 | |
| Front brake hose holder and steering knuckle | M6 | 7 | 0.7 | 5.1 | |
| Front brake hose holder and front upper arm | M6 | 7 | 0.7 | 5.1 | |
| Front brake pad holding bolt | M8 | 17 | 1.7 | 12 | |
| Front brake caliper bleed screw | M6 | 5 | 0.5 | 3.6 | |
| Rear upper arm and frame | M10 | 45 | 4.5 | 32 |  |
| Rear lower arm and frame | M10 | 45 | 4.5 | 32 |  |
| Rear knuckle and rear upper arm | M10 | 45 | 4.5 | 32 | |
| Rear knuckle and rear lower arm | M10 | 45 | 4.5 | 32 | |
| Rear shock absorber and frame | M12 | 45 | 4.5 | 32 | |
| Rear shock absorber and rear lower arm | M12 | 45 | 4.5 | 32 | |
| Rear arm protector and rear lower arm | M6 | 7 | 0.7 | 5.1 | |
| Stabilizer holder and frame | M8 | 42 | 4.2 | 30 | |

| Part to be tightened | Thread size | Tightening torque | | | Remarks |
|---|---|---|---|---|---|
| | | Nm | m · kg | ft · lb | |
| Stabilizer joint and stabilizer | M10 | 60 | 6.0 | 43 | |
| Stabilizer joint and rear lower arm | M10 | 60 | 6.0 | 43 | |
| Rear wheel and rear wheel hub | M10 | 55 | 5.5 | 40 | |
| Rear wheel hub and constant velocity joint | M20 | 350 | 35.0 | 250 | Stake. |
| Rear knuckle and brake disc guard | M6 | 7 | 0.7 | 5.1 | |
| Rear knuckle and plate (for models equipped with cast wheels) | M6 | 7 | 0.7 | 5.1 | |
| Rear brake caliper and rear knuckle | M10 | 48 | 4.8 | 35 | |
| Brake pipe and brake master cylinder | M10 | 19 | 1.9 | 13 | |
| Brake pipe and brake hose joint | M10 | 19 | 1.9 | 13 | |
| Brake hose joint and frame | M6 | 7 | 0.7 | 5.1 | |
| Brake master cylinder and pedal assembly | M8 | 16 | 1.6 | 11 | |
| Pedal assembly and frame | M8 | 16 | 1.6 | 11 | |
| Secondary brake master cylinder kit stopper bolt | M6 | 9 | 0.9 | 6.5 | |
| Brake rod locknut | M8 | 17 | 1.7 | 12 | |
| Rear brake disc and rear wheel hub | M8 | 30 | 3.0 | 22 |  |
| Rear brake pad holding bolt | M8 | 17 | 1.7 | 12 | |
| Rear brake caliper bleed screw | M6 | 5 | 0.5 | 3.6 | |
| Rear brake hose union bolt | M10 | 27 | 2.7 | 19 | |
| Parking brake lever assembly and parking brake lever assembly bracket | M6 | 13 | 1.3 | 9.4 | |
| Parking brake lever assembly bracket and frame | M10 | 32 | 3.2 | 23 | |
| Parking brake cable holder and frame | M6 | 7 | 0.7 | 5.1 | |
| Parking brake assembly and final drive gear assembly | M10 | 40 | 4.0 | 29 | |
| Parking brake disc and final drive gear assembly | M6 | 10 | 1.0 | 7.2 | |
| Parking brake pad holding bolt | M10 | 17 | 1.7 | 12 |  |
| Parking brake arm nut | M8 | 17 | 1.7 | 12 | |
| Parking brake cable holder | M6 | 12 | 1.2 | 8.7 | |
| Differential gear case assembly and frame | M10 | 55 | 5.5 | 40 | |
| Differential gear case filler plug | M14 | 23 | 2.3 | 17 | |
| Differential gear case drain plug | M10 | 10 | 1.0 | 7.2 | |
| Universal joint yoke and differential drive pinion gear | M14 | 62 | 6.2 | 45 | |
| Differential gear motor and differential gear case cover | M6 | 11 | 1.1 | 8.0 | |
| Differential gear case cover and differential gear case | M8 | 24 | 2.4 | 17 | |
| Final drive gear assembly and frame | M10 | 70 | 7.0 | 50 | |
| Final gear case filler plug | M20 | 23 | 2.3 | 17 | |
| Final gear case drain plug | M10 | 20 | 2.0 | 14 | |
| Final drive pinion gear bearing housing and final gear case | M8 | 32 | 3.2 | 23 | |

| Part to be tightened | Thread size | Tightening torque | | | Remarks |
|---|---|---|---|---|---|
| | | Nm | m · kg | ft · lb | |
| Ring gear bearing housing and final gear case | M10 | 40 | 4.0 | 29 | |
| | M8 | 23 | 2.3 | 17 | |
| Ring gear stopper nut | M8 | 16 | 1.6 | 11 | |
| Bearing retainer and final gear pinion gear bearing housing | M65 | 170 | 17.0 | 125 | Left-hand thread. |
| Drive shaft coupling gear and final drive pinion gear | M12 | 80 | 8.0 | 58 | |

EBS00022

## HOW TO USE THE CONVERSION TABLE

All specification data in this manual are listed in SI and METRIC UNITS.
Use this table to convert METRIC unit data to IMPERIAL unit data.

Ex.

| METRIC | | MULTIPLIER | | IMPERIAL |
|---|---|---|---|---|
| ** mm | × | 0.03937 | = | ** in |
| 2 mm | × | 0.03937 | = | 0.08 in |

CONVERSION TABLE

| METRIC TO IMPERIAL | | | |
|---|---|---|---|
| | Metric unit | Multiplier | Imperial unit |
| Torque | m · kg<br>m · kg<br>cm · kg<br>cm · kg | 7.233<br>86.794<br>0.0723<br>0.8679 | ft · lb<br>in · lb<br>ft · lb<br>in · lb |
| Weight | kg<br>g | 2.205<br>0.03527 | lb<br>oz |
| Speed | km/hr | 0.6214 | mph |
| Distance | km<br>m<br>m<br>cm<br>mm | 0.6214<br>3.281<br>1.094<br>0.3937<br>0.03937 | mi<br>ft<br>yd<br>in<br>in |
| Volume/ Capacity | cc ($cm^3$)<br>cc ($cm^3$)<br>lt (liter)<br>lt (liter) | 0.03527<br>0.06102<br>0.8799<br>0.2199 | oz (IMP liq.)<br>cu · in<br>qt (IMP liq.)<br>gal (IMP liq.) |
| Misc. | kg/mm<br>$kg/cm^2$<br>Centigrade (°C) | 55.997<br>14.2234<br>9/5+32 | lb/in<br>psi ($lb/in^2$)<br>Fahrenheit (°F) |

EBS00023

## GENERAL TIGHTENING TORQUE SPECIFICATIONS

This chart specifies tightening torques for standard fasteners with a standard ISO thread pitch. Tightening torque specifications for special components or assemblies are provided for each chapter of this manual. To avoid warpage, tighten multi-fastener assemblies in a crisscross pattern and progressive stages until the specified tightening torque is reached. Unless otherwise specified, tightening torque specifications require clean, dry threads. Components should be at room temperature.
A: Distance between flats
B: Outside thread diameter

| A (nut) | B (bolt) | General tightening torques | | |
|---|---|---|---|---|
| | | Nm | m · kg | ft · lb |
| 10 mm | 6 mm | 6 | 0.6 | 4.3 |
| 12 mm | 8 mm | 15 | 1.5 | 11 |
| 14 mm | 10 mm | 30 | 3.0 | 22 |
| 17 mm | 12 mm | 55 | 5.5 | 40 |
| 19 mm | 14 mm | 85 | 8.5 | 61 |
| 22 mm | 16 mm | 130 | 13.0 | 94 |

EBS00024

# LUBRICATION POINTS AND LUBRICANT TYPES

## ENGINE

| Lubrication points | Lubricant |
|---|---|
| Oil seal lips | LS |
| Bearings | E |
| O-ring | LS |
| Cylinder head bolts | M |
| Crankshaft pin | E |
| Connecting rod big end thrust surface | E |
| Crankshaft sprocket | M |
| Inner race (crankshaft) | E |
| Buffer boss (crankshaft) | E |
| Crankshaft seal | E |
| Piston pin | E |
| Piston and ring grooves | E |
| Valve stems (intake and exhaust) | M |
| Valve stem ends (intake and exhaust) | M |
| Rocker arm shafts | E |
| Camshaft lobes | M |
| Decompressor lever pin | E |
| Decompressor lever spring | E |
| Rocker arms (intake and exhaust) | M |
| Oil pump shaft | E |
| O-ring (oil filter cartridge) | LS |
| Water pump impeller shaft | M |
| Dipstick mating surface | E |
| Starter idler gear inner surface | E |
| Starter idler gear shaft | E |
| Starter wheel gear | E |
| Torque limiter | E |
| Clutch housing shaft end | LS |
| Clutch carrier assembly | E |
| One-way clutch bearing | E |
| Clutch dog and middle drive gear | M |
| Reverse idle gear shaft | E |
| Middle driven shaft splines | M |
| Shift drum | E |
| Shift forks and shift fork guide bar | E |
| Ball (shift drum stopper) | E |
| Shift lever 2 inner surface | LS |

EBS00029

# PERIODIC CHECKS AND ADJUSTMENTS

## INTRODUCTION

This chapter includes all information necessary to perform recommended checks and adjustments. These preventive maintenance procedures, if followed, will ensure more reliable vehicle operation and a longer service life. The need for costly overhaul work will be greatly reduced. This information applies to vehicles already in service as well as to new vehicles that are being prepared for sale. All service technicians should be familiar with this entire chapter.

## PERIODIC MAINTENANCE CHART FOR THE EMISSION CONTROL SYSTEM

- For vehicles not equipped with an odometer or hour meter, follow the month maintenance intervals.
- For vehicles equipped with an odometer or an hour meter, follow the km (mi) or hours maintenance intervals. However, keep in mind that if the vehicle isn't used for a long period of time, the month maintenance intervals should be followed.
- Items marked with an asterisk should be performed by a Yamaha dealer as they require special tools, data and technical skills.



| ITEM | ROUTINE | Whichever comes first ⇨ | | INITIAL | | | EVERY | |
|---|---|---|---|---|---|---|---|---|
| | | | month | 1 | 3 | 6 | 6 | 12 |
| | | | km (mi) | 320 (200) | 1,200 (750) | 2,400 (1,500) | 2,400 (1,500) | 4,800 (3,000) |
| | | | hours | 20 | 75 | 150 | 150 | 300 |
| **Fuel line*** | • Check fuel hose for cracks or damage.<br>• Replace if necessary. | | | | | ○ | ○ | ○ |
| **Valves*** | • Check valve clearance.<br>• Adjust if necessary. | | | ○ | | ○ | ○ | ○ |
| **Spark plug** | • Check condition.<br>• Adjust gap and clean.<br>• Replace if necessary. | | | ○ | ○ | ○ | ○ | ○ |
| **Crankcase breather system*** | • Check breather hose for cracks or damage.<br>• Replace if necessary. | | | | | ○ | ○ | ○ |
| **Exhaust system*** | • Check for leakage.<br>• Tighten if necessary.<br>• Replace gasket(s) if necessary. | | | | | ○ | ○ | ○ |
| **Spark arrester** | • Clean. | | | | | ○ | ○ | ○ |

# GENERAL MAINTENANCE AND LUBRICATION CHART

| ITEM | ROUTINE | Whichever comes first ⇨ | | INITIAL | | | EVERY | |
|---|---|---|---|---|---|---|---|---|
| | | | month | 1 | 3 | 6 | 6 | 12 |
| | | | km (mi) | 320 (200) | 1,200 (750) | 2,400 (1,500) | 2,400 (1,500) | 4,800 (3,000) |
| | | | hours | 20 | 75 | 150 | 150 | 300 |
| **Cooling system** | • Check coolant leakage.<br>• Repair if necessary.<br>• Replace coolant every 24 months. | | | ○ | ○ | ○ | ○ | ○ |
| **Air filter element** | • Clean.<br>• Replace if necessary. | | | Every 20–40 hours (More often in wet or dusty areas.) | | | | |
| **Engine oil** | • Replace (warm engine before draining). | | | ○ | | ○ | ○ | ○ |
| **Engine oil filter cartridge** | • Replace. | | | ○ | | ○ | | ○ |
| **Final gear oil**<br>**Differential gear oil** | • Check oil level/oil leakage.<br>• Replace. | | | ○ | | | | ○ |
| **Front brake*** | • Check operation/brake pad wear/fluid leakage<br>• Correct if necessary. Replace pads if worn to the limit. | | | ○ | ○ | ○ | ○ | ○ |
| **Rear brake*** | • Check operation/brake pad wear/fluid leakage<br>• Correct if necessary. Replace pads if worn to the limit. | | | ○ | ○ | ○ | ○ | ○ |
| **Parking brake*** | • Check operation and free play/brake pad wear.<br>• Correct if necessary. Replace pads if worn to the limit. | | | ○ | ○ | ○ | ○ | ○ |
| **Accelerator pedal*** | • Check operation and free play. | | | ○ | ○ | ○ | ○ | ○ |
| **V-belt*** | • Check operation.<br>• Check for wear, cracks, or damage. | | | ○ | | | ○ | ○ |
| **Wheels*** | • Check balance/damage/runout.<br>• Repair if necessary. | | | ○ | | ○ | ○ | ○ |
| **Wheel bearings*** | • Check bearing assemblies for looseness/damage.<br>• Replace if damaged. | | | ○ | | ○ | ○ | ○ |
| **Front and rear suspension*** | • Check operation and for leakage.<br>• Correct if necessary. | | | | | ○ | | ○ |
| **Steering system*** | • Check operation and for looseness/Replace if damaged.<br>• Check toe-in/Adjust if necessary. | | | ○ | ○ | ○ | ○ | ○ |
| **Rear upper and lower knuckle pivots*** | • Lubricate with lithium-soap-based grease. | | | | | ○ | ○ | ○ |
| **Drive shaft universal joint*** | • Lubricate with lithium-soap-based grease. | | | | | ○ | ○ | ○ |
| **Engine mount*** | • Check for cracks or damage.<br>• Check bolt tightness. | | | | | ○ | ○ | ○ |
| **Front and rear axle boots*** | • Check operation.<br>• Replace if damaged. | | | ○ | | | | ○ |
| **Stabilizer bushings*** | • Check for cracks or damage. | | | | | ○ | ○ | ○ |
| **Fittings and fasteners*** | • Check all chassis fittings and fasteners.<br>• Correct if necessary. | | | ○ | ○ | ○ | ○ | ○ |

**Recommended brake fluid: DOT 4**

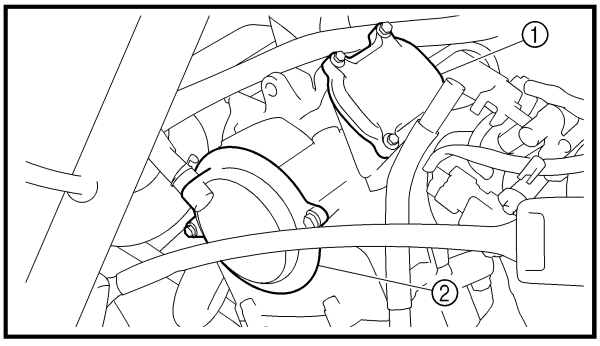
**Brake fluid replacement:**

- Replace the brake fluid when disassembling the master cylinder or caliper.
- Check the brake fluid level regularly and add fluid as required.
- Replace the oil seals on the inner parts of the master cylinder and caliper every two years.
- Replace the brake hoses every four years, or if cracked or damaged.

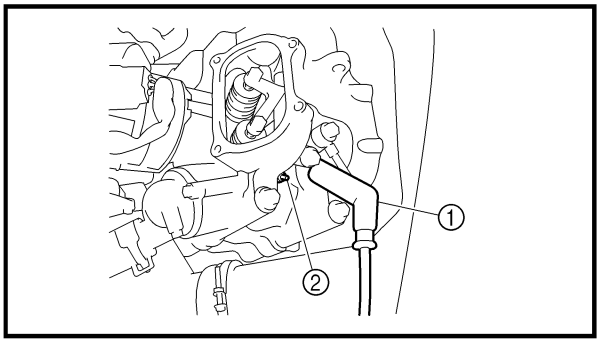
EAS00049
# ENGINE
## ADJUSTING THE VALVE CLEARANCE
The following procedure applies to all of the valves.

**TIP:**
- Valve clearance adjustment should be made on a cold engine, at room temperature.
- When the valve clearance is to be measured or adjusted, the piston must be at top dead center (TDC) on the compression stroke.

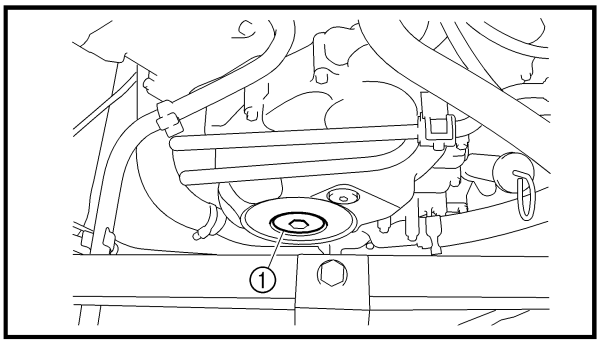
1. Remove:
   - seats
   - rear console
     Refer to “SEATS, REAR CONSOLE AND INSTRUMENT PANELS” in chapter 8.
   - center protector
     Refer to “PANELS AND FRONT CONSOLE” in chapter 8.




2. Remove:
   - intake tappet cover ①
   - exhaust tappet cover
   - camshaft sprocket cover ②



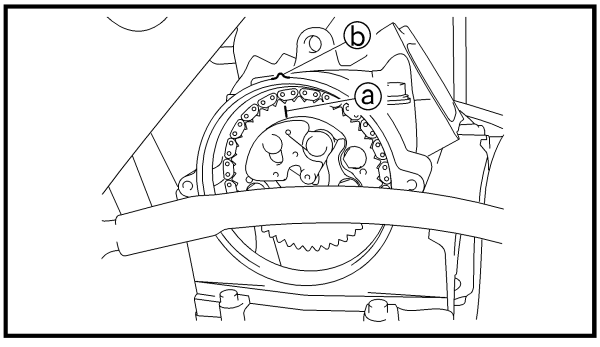

3. Disconnect:
   - spark plug cap ①
4. Remove:
   - spark plug ②




5. Remove:
   • crankshaft end accessing screw ①

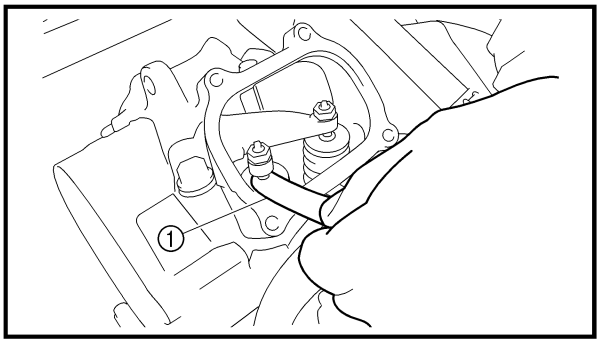
6. Measure:
   • valve clearance
   Out of specification → Adjust.

**Valve clearance (cold)**
**Intake valve**
**0.09 ~ 0.13 mm**
**(0.0035 ~ 0.0051 in)**
**Exhaust valve**
**0.16 ~ 0.20 mm**
**(0.0063 ~ 0.0079 in)**




▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼

a. Turn the crankshaft counterclockwise.
b. When the piston is at the top dead center (TDC) on the compression stroke, align the "I" mark ⓐ on the camshaft sprocket with the stationary pointer ⓑ on the cylinder head.

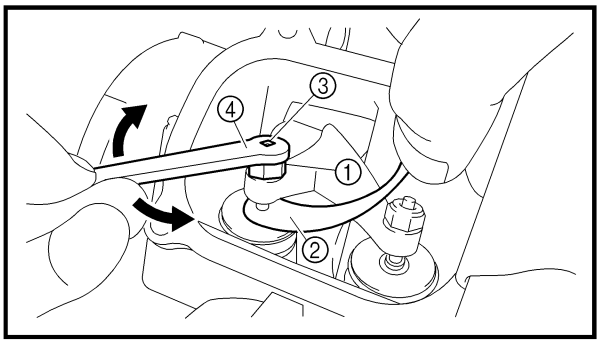



c. Measure the valve clearance with a thickness gauge ①.

**Thickness gauge**
**90890-03180**
**Feeler gauge set**
**YU-26900-9**

Out of specification → Adjust.

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7. Adjust:
   • valve clearance

▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼▼

a. Loosen the locknut ①.
b. Insert a thickness gauge ② between the adjusting screw and the valve tip.
c. Turn the adjusting screw ③ with the tappet adjusting tool ④ until the specified valve clearance is obtained.



**Tappet adjusting tool**
**90890-01311**
**Six piece tappet set**
**YM-A5970**

d. Hold the adjusting screw to prevent it from moving and tighten the locknut to the specified torque.



**Locknut**
**14 Nm (1.4 m · kg, 10 ft · lb)**

e. Measure the valve clearance again.
f. If the valve clearance is still out of specification, repeat all of the valve clearance adjustment steps until the specified clearance is obtained.

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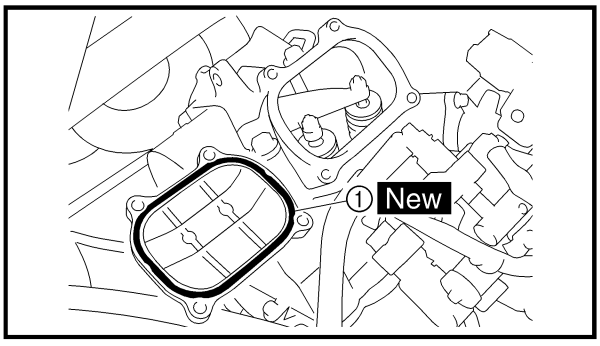
8. Install:
   • crankshaft end accessing screw

**10 Nm (1.0 m · kg, 7.2 ft · lb)**

9. Install:
   • spark plug

**13 Nm (1.3 m · kg, 9.4 ft · lb)**

10. Connect:
   • spark plug cap




11. Install:
- O-ring New
- camshaft sprocket cover

10 Nm (1.0 m · kg, 7.2 ft · lb)

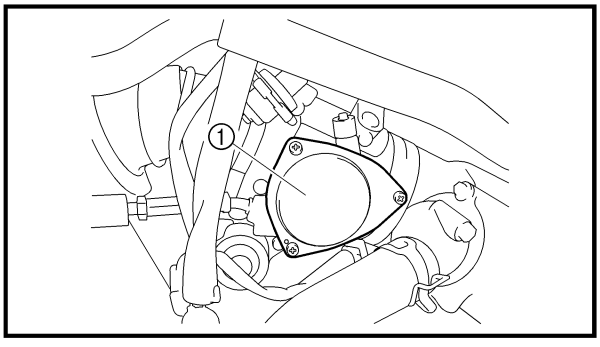
- O-ring ① New
- intake tappet cover

10 Nm (1.0 m · kg, 7.2 ft · lb)

- O-ring New
- exhaust tappet cover

10 Nm (1.0 m · kg, 7.2 ft · lb)

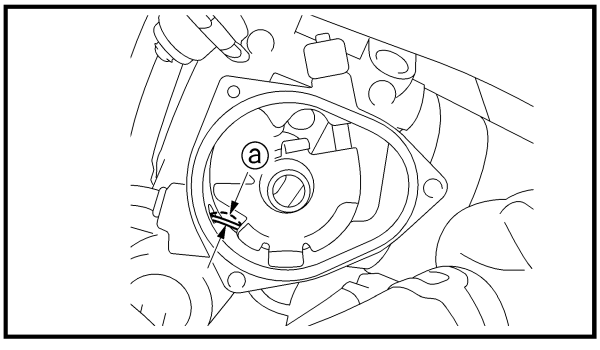
12. Install:
- center protector
  Refer to “PANELS AND FRONT CONSOLE” in chapter 8.
- rear console
- seats
  Refer to “SEATS, REAR CONSOLE AND INSTRUMENT PANELS” in chapter 8.



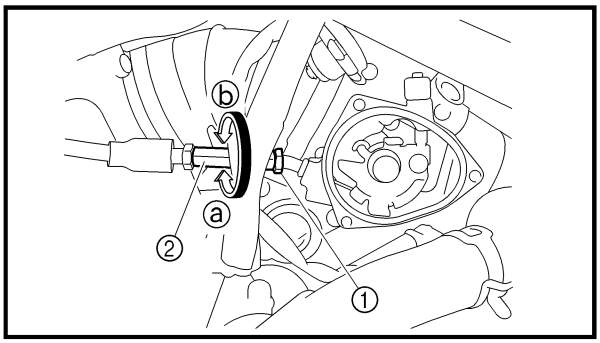

**ADJUSTING THE THROTTLE CABLE**

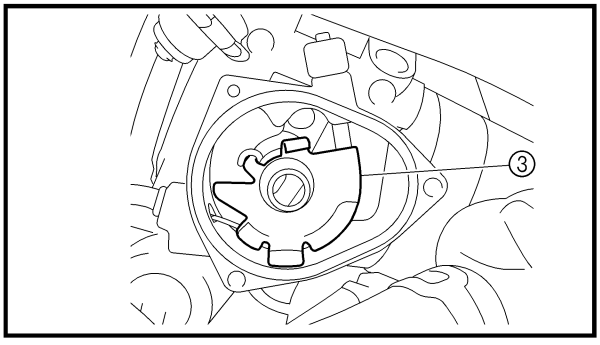
1. Remove:
- seats
- rear console
  Refer to “SEATS, ENCLOSURE, HOOD AND CARGO BED” in chapter 8.

2. Remove:
- throttle valve cover ①




3. Check:
- throttle cable ⓐ
  Slack → Remove the slack.




4. Adjust:
   • throttle cable

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a. Loosen the locknut ①.
b. Turn the adjusting nut ② in direction ⓐ or ⓑ until the correct free play is obtained.

| | |
|---|---|
| **Direction ⓐ** | **Free play is increased.** |
| **Direction ⓑ** | **Free play is decreased.** |

c. Tighten the locknut.

**TIP:**
After adjusting the throttle cable, depress the accelerator pedal a few times and make sure that the throttle valve ③ closes completely after releasing the accelerator pedal.

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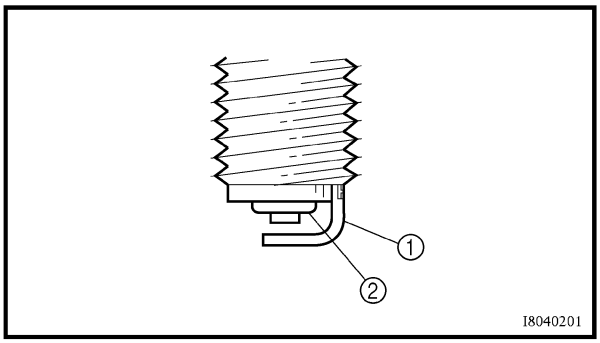
5. Install:
   • throttle valve cover
6. Install:
   • rear console
   • seats
   Refer to “SEATS, REAR CONSOLE AND INSTRUMENT PANELS” in chapter 8.

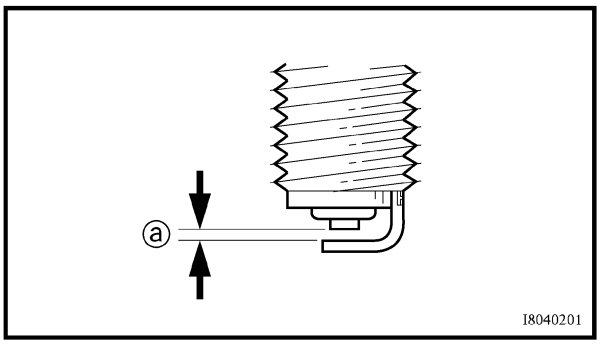
EBS00057

## CHECKING THE SPARK PLUG

1. Remove:
   • seats
   • rear console
   Refer to “SEATS, ENCLOSURE, HOOD AND CARGO BED” in chapter 8.
2. Disconnect:
   • spark plug cap
3. Remove:
   • spark plug
4. Check:
   • spark plug type
   Incorrect → Change.

| | |
|---|---|
| | **Standard spark plug**<br>**CPR7EA-9/NGK** |




5. Check:
   - electrode ①
     Wear/damage → Replace.
   - insulator ②
     Abnormal color → Replace.
     Normal color is a medium-to-light tan color.
6. Clean:
   - spark plug
     (with a spark plug cleaner or wire brush)

7. Measure:
   - spark plug gap ⓐ
     Use a wire gauge or thickness gauge.
     Out of specification → Regap.

| | |
|---|---|
| | **Spark plug gap**<br>**0.8 ~ 0.9 mm (0.031 ~ 0.035 in)** |

8. Tighten:
   - spark plug — **13 Nm (1.3 m · kg, 9.4 ft · lb)**

**TIP:**
Before installing a spark plug, clean the gasket surface and plug surface.

9. Connect:
   - spark plug cap
10. Install:
    - rear console
    - seats
      Refer to “SEATS, ENCLOSURE, HOOD AND CARGO BED” in chapter 8.

EBS00058

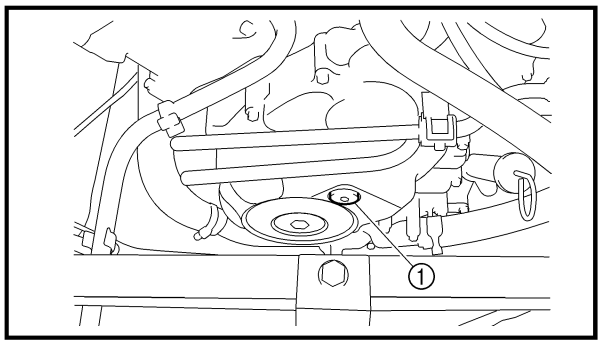
## CHECKING THE IGNITION TIMING

**TIP:**

Throttle cable free play should be adjusted properly before checking the ignition timing.

1. Remove:
   - seats
   - rear console
     Refer to “SEATS, REAR CONSOLE AND INSTRUMENT PANELS” in chapter 8.
   - fuel tank
     Refer to “FUEL PUMP AND FUEL TANK” in chapter 6.
2. Attach:
   - tachometer
   - timing light
     (to spark plug lead)

**Digital tachometer**
**90890-06760, YU-39951-B**
**Timing light**
**90890-03141**
**Inductive clamp timing light**
**YU-03141**



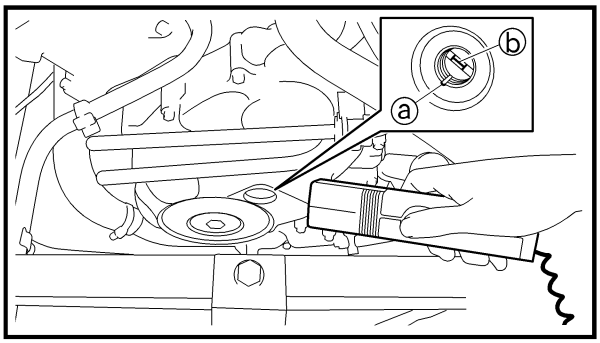

3. Check:
   - ignition timing

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a. Warm up the engine and keep it at the specified speed.

**Engine speed**
**1,550 ~ 1,650 r/min**

b. Remove the timing mark accessing screw ①.




c. Visually check the stationary pointer ⓐ to verify it is within the required firing range ⓑ indicated on the AC magneto rotor.
Incorrect firing range → Check the ignition system.

**TIP:**
When checking the ignition timing, make sure that the timing light cord does not come in contact with the exhaust muffler.

d. Install the timing mark accessing screw.

| | |
|---|---|
| | **Timing mark accessing screw 6 Nm (0.6 m · kg, 4.3 ft · lb)** |

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4. Detach:
   - timing light
   - tachometer
5. Install:
   - fuel tank
     Refer to “FUEL PUMP AND FUEL TANK” in chapter 6.
   - rear console
   - seats
     Refer to “SEATS, REAR CONSOLE AND INSTRUMENT PANELS” in chapter 8.

EBS00061

## MEASURING THE COMPRESSION PRESSURE

**TIP:**
Insufficient compression pressure will result in a loss of performance.

1. Measure:
   - valve clearance
     Out of specification → Adjust.
     Refer to “ADJUSTING THE VALVE CLEARANCE”.
2. Start the engine, warm it up for several minutes, and then turn it off.
3. Remove:
   - seats
   - rear console
     Refer to “SEATS, REAR CONSOLE AND INSTRUMENT PANELS” in chapter 8.